

Conjecture 00000002805 is false:
the Schilder rate does not vanish on the Cameron–Martin unit ball

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1 The conjecture

Conjecture 00000002805 (English text, verbatim): “Definition: Schilder’s theorem: the LDP on the path space of Brownian motion. Conjecture: Cameron–Martin rigidity of the Schilder rate: the rate function vanishes if and only if the path lies in the Cameron–Martin unit ball, and the spectral distribution of the rate on the sphere is an explicit chi-square-type law.” The Chinese text says the same.

Reading. The conjecture is a conjunction. We refute its first clause, the “Cameron–Martin rigidity” named in the title:

$$I(\omega) = 0 \iff \omega \text{ lies in the Cameron–Martin unit ball,} \quad (*)$$

where I is the rate function of Schilder’s theorem. We refute $(*)$ for the closed unit ball $\{\|\omega\|_H \leq 1\}$ and for the open unit ball $\{\|\omega\|_H < 1\}$, in every dimension $d \geq 1$ and for every time horizon $T > 0$. The second clause (the “spectral distribution on the sphere”) is not interpreted and not used; refuting one conjunct refutes the conjunction.

2 Definitions (Schilder’s theorem)

We use the statement of Schilder’s theorem as given on Wikipedia (“Schilder’s theorem”, retrieved 2026-10-05). Let B be a standard Brownian motion in $\mathbb{R}^d$ started at 0, and $C_0 = C_0([0, T]; \mathbb{R}^d)$ the continuous paths ω on $[0, T]$ with $\omega(0) = 0$. For $\varepsilon > 0$ the laws of $\sqrt{\varepsilon}B$ satisfy the large deviation principle on C_0 “with good rate function $I : C_0 \rightarrow \mathbb{R} \cup \{+\infty\}$ given by $I(\omega) = \frac{1}{2} \int_0^T |\dot{\omega}(t)|^2 dt$ if ω is absolutely continuous, and $I(\omega) = +\infty$ otherwise”. The Cameron–Martin space is “the subspace of absolutely continuous functions whose derivative is in L^2 ”. Its norm is $\|\omega\|_H = (\int_0^T |\dot{\omega}(t)|^2 dt)^{1/2}$, so $I(\omega) = \frac{1}{2} \|\omega\|_H^2$ on H .

Definition 1 (as in Lean). $\mathbb{R}^d$ is `EuclideanSpace ℝ (Fin d)`; paths are functions $\omega : \mathbb{R} \rightarrow \mathbb{R}^d$ looked at on $[0, T]$.

- `PathSpace d T`: ω continuous on $[0, T]$ and $\omega(0) = 0$.
- `CameronMartin d T`: $\omega(0) = 0$, ω absolutely continuous on $[0, T]$ (`Mathlib’s AbsolutelyContinuousOnInterval`), and $\dot{\omega} = \text{deriv } \omega$ in $L^2([0, T]; \mathbb{R}^d)$ (`MemLp (deriv omega) 2`).

- `cmNormSq T omega` = $\int_{[0,T]} \|\dot{\omega}(t)\|^2 dt \in [0, \infty]$ (lower Lebesgue integral), i.e. $\|\omega\|_H^2$.
- `cmClosedUnitBall`, `cmOpenUnitBall`: $\omega \in H$ with $\|\omega\|_H^2 \leq 1$, resp. < 1 .
- `schilderRate d T omega` = $\frac{1}{2} \|\omega\|_H^2$ if ω is absolutely continuous on $[0, T]$, and $+\infty$ otherwise (values in $[0, \infty]$).

Using the lower integral makes $I(\omega) = +\infty$ automatically when ω is absolutely continuous but $\dot{\omega} \notin L^2$, which agrees with the textbook rate.

3 The counterexample

Let e_0 be the first standard basis vector of $\mathbb{R}^d$ ($d \geq 1$), put $w = \frac{1}{2\sqrt{T}}e_0$ and $\omega_0(t) = t w$.

Theorem 2. $\omega_0 \in C_0$, $\omega_0 \in H$, $\|\omega_0\|_H^2 = \frac{1}{4}$ and $I(\omega_0) = \frac{1}{8}$. Hence ω_0 lies in the open (and so in the closed) Cameron–Martin unit ball but $I(\omega_0) \neq 0$, and $(*)$ fails for both balls.

Proof. ω_0 is linear, so it is continuous, $\omega_0(0) = 0$, it is C^1 and hence absolutely continuous on $[0, T]$, and $\dot{\omega}_0(t) = w$ for every t . A constant is in L^2 of the finite measure space $[0, T]$, so $\omega_0 \in H$. Since $\|e_0\| = 1$, $\|w\| = \frac{1}{2\sqrt{T}}$ and

$$\|\omega_0\|_H^2 = \int_0^T \frac{1}{4T} dt = \frac{1}{4} < 1, \quad I(\omega_0) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8} \neq 0.$$

If $(*)$ held for all $\omega \in C_0$, the implication “in the ball $\Rightarrow I = 0$ ” applied to ω_0 would give $I(\omega_0) = 0$, a contradiction. $\square$

Remark (what is true). $I(\omega) = \frac{1}{2}\|\omega\|_H^2$ vanishes exactly at the zero path, while the Cameron–Martin unit ball is the sublevel set $\{I \leq \frac{1}{2}\}$. This remark is not formalized and is not used.

4 Lean formalization

File `lean/Conjecture2805/Basic.lean` (148 lines, only import `Mathlib`), namespace `C2805`. Objects: `PathSpace`, `CameronMartin`, `cmNormSq`, `cmClosedUnitBall`, `cmOpenUnitBall`, `schilderRate` as above; `dir d T` = $\frac{1}{2\sqrt{T}}e_0$; `straightPath d T` = $t \mapsto t \cdot \text{dir } d \text{ } T$. Lemmas: `straightPath_mem_pathSpace`, `straightPath_mem_cameronMartin` (absolute continuity from `ContDiffOn.absolutelyContinuousOnInterval`), `cmNormSq_straightPath` (= $1/4$), `schilderRate_straightPath` (= $1/8$), `straightPath_mem_open_ball`, `straightPath_mem_closed_ball`, `schilderRate_straightPath_ne_zero`. Main theorems, for every d with `[NeZero d]` and every $T > 0$:

- `schilder_rigidity_false`: `not (forall omega in PathSpace d T, (schilderRate d T omega = 0 <-> omega in cmClosedUnitBall d T))`;
- `schilder_rigidity_false_open`: the same with `cmOpenUnitBall`.

`#print` axioms for both main theorems and for `schilderRate_straightPath` gives `[propext, Classical.choice, Quot.sound]`; no sorry, no native_decide. Checked with Lean 4.33.1 and Mathlib v4.33.1 (lake build).

5 Relation to the earlier submission

An earlier submission (PR #277, orionsheep) gave the same counterexample $\omega(t) = t/2$ but was closed. The reviewer wrote that its main theorem “is entirely vacuous $\mathbb{N}$ trivia ($\forall c > 0, 0 < c \cdot c, 1 \leq 2, 1 \neq 0, 2 = 1 \cdot 2, 1 \neq 2$); the Schilder rate, Cameron–Martin space, sphere, and chi-square law never appear in Lean”. This submission defines in Lean the path space $C_0([0, T]; \mathbb{R}^d)$, the Cameron–Martin space (absolutely continuous, derivative in L^2), the Cameron–Martin norm and both unit balls, and Schilder’s rate function with the value $+\infty$ off the absolutely continuous paths; it proves the counterexample for every $d \geq 1$ and $T > 0$. Unlike PR #277 it makes no claim about the second clause (the sphere and the chi-square law), which is left uninterpreted.

Source

Wikipedia, “Schilder’s theorem”, https://en.wikipedia.org/wiki/Schilder%27s_theorem (definitions quoted above).